

CSE7000 - INTERNSHIP

A Presentation on

AI-Based Smart Exam Proctoring and Malpractice Detection

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About Company or Organization

- **The AI Center of Excellence, Accelerated by NVIDIA**, is an initiative focused on promoting practical learning and innovation in Artificial Intelligence and emerging technologies. It provides an environment for students and learners to gain hands-on experience in areas such as Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and Generative AI.
- As part of the internship, practical knowledge was gained through the development of an AI-based project, applying concepts of Computer Vision and Machine Learning to a real-world problem. The internship provided exposure to AI technologies, software development practices, problem-solving, and the implementation of an end-to-end AI solution.

• Internship and Work Environment

- During the internship at the AI Center of Excellence, Accelerated by NVIDIA, the focus was on applying AI concepts through practical project development.
- Worked on an AI-based real-world problem.
- Gained practical experience in Computer Vision and Machine Learning.
- Developed an AI-based Proctoring System .
- Applied object detection and tracking techniques using YOLOv8.
- Worked with Python, Flask, MongoDB Atlas, HTML, CSS, and JavaScript.
- Gained experience in integrating AI models with a complete software application.
- Developed skills in problem-solving, system design, implementation, and testing.



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Working domain or the technology

- The project was developed in the domain of Artificial Intelligence, Computer Vision, and Online Examination Security. The main technologies used include Python, Flask, OpenCV, YOLOv8, MediaPipe Face Mesh, HTML, CSS, JavaScript, and SQLite.
- YOLOv8 was used for real-time mobile phone and multiple-person detection, while MediaPipe Face Mesh was used for face detection and head pose estimation. Flask handled the backend, SQLite stored examination logs and evidence, and HTML, CSS, and JavaScript were used to develop the student and admin interfaces.



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Objectives of the work

- Ensure fair and cheat-free online examinations by monitoring candidates automatically.
- Detect suspicious activities such as multiple faces, absence from screen, or unusual movements.
- Use face recognition and authentication to verify candidate identity.
- Enable real-time alerts and flagging of violations during exams.
- Reduce the need for human invigilators, saving time and operational costs.
- Maintain exam integrity and credibility in remote learning environments.
- Provide recorded sessions and reports for post-exam review and audit.
- Support scalable exam management for large numbers of students simultaneously.



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Methodology Used / Course Description

1. Requirement Analysis

Identified the main requirements such as student login, admin login, online test, question bank, monitoring, and result generation.

2. Database Development

Created the SQLite database to store students, questions, examination attempts, answers, results, events, and evidence.

3. Backend Development

Developed the Flask backend and connected it with the database to handle examination and student information.

4. Test Interface

Created the student test page with questions, options, timer, question palette, navigation, and submission features.



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5. Backend Development

Developed the backend using Python and Flask to handle the examination process, database communication, student responses, and monitoring functions.

6. Database Management

Used SQLite to store student details, questions, examination attempts, answers, results, proctoring events, and captured evidence.

7. Monitoring and Alerts

Monitored students through the webcam and microphone and recorded suspicious activities such as no face, multiple faces, mobile phone detection, tab switching, and fullscreen exit.

8. Testing & Evaluation

Tested the login, examination, monitoring, evaluation, risk score, event logging, and administrator modules to check that the main system functions work correctly.



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Results

ONLINE APTITUDE TEST

Student: student001

Time Remaining: 59:11

Question 45 of 50

A can do a piece of work in 20 days and B can do it in 20 days. How long will they take if they work together?

☐ 10.00 days

☐ 20.00 days

☐ 10.00 days

☐ 12.00 days

Previous

Save & Next

Proctoring: Active

Face: ✓ Detected

Microphone: Active

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35
36	37	38	39	40
41	42	43	44	45
46	47	48	49	50

Submit Exam

Figure : Online Test Interface

Attempt Details

Student ID: student001

Exam Information

Student Name

Demo Student

Started

2026-08-09 23:30:09

Completed

2026-08-09 23:31:13

Status

COMPLETED

Final Score

0 / 50

Percentage

0.0%

Proctoring Summary

TOTAL RISK SCORE

50

Medium Risk

Proctoring Events Log

7 Events Recorded

TIME	EVENT TYPE	DESCRIPTION	SEVERITY	EVIDENCE
23:31:14	NO_FACE	No face detected in frame	Medium (10)	View Image
23:31:13	FULLSCREEN_EXIT	Browser event: FULLSCREEN_EXIT	Low (5)	-
23:31:13	TAB_SWITCH	Browser event: TAB_SWITCH	Low (5)	-
23:31:03	NO_FACE	No face detected in frame	Medium (10)	View Image
23:31:00	TAB_SWITCH	Browser event: TAB_SWITCH	Low (5)	-
23:30:29	TAB_SWITCH	Browser event: TAB_SWITCH	Low (5)	-
23:30:27	NO_FACE	No face detected in frame	Medium (10)	View Image

Figure 1.2: Proctoring Attempt Details



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Conclusion

- The AI-Based Smart Exam Proctoring and Malpractice Detection System was developed to conduct online tests with features such as random questions, webcam and microphone monitoring, mobile phone detection, browser monitoring, and automatic evaluation.
- The system records suspicious activities, calculates a risk score, and provides examination results and evidence for administrator review. The project also provided practical experience in using Python, Flask, OpenCV, YOLOv8, JavaScript, and SQLite.



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Q&A



Thank you !!



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